Lars Schöbitz

Open Science Specialist, ETH Zurich - Global Health Engineering. Director & Founder, Lars Schöbitz GmbH.
lars@lse.de · lschoebitz@ethz.ch · lse.de · github.com/larnsce · ORCID 0000-0003-2196-5015 · linkedin.com/in/larsschoebitz

Environmental engineer and data scientist working in the global water, sanitation and hygiene (WASH) sector. Open Science Specialist at ETH Zurich, where I help researchers make their work reproducible and their data open. I founded and lead openwashdata, teach data science with R, and write and manage the grants that fund this work.
This CV is rendered from the aboutme data package; the online version and a one-page CV are at www.lse.de/cv.html.

Technical skills

- R / tidyverse
- Quarto / R Markdown
- Git / GitHub
- R Shiny
- Reproducible research
- SQL
- HTML / CSS

Languages

- German (native)
- English (fluent)
- Swedish
- French

WORK EXPERIENCE

Since 2021 — **Open Science Specialist**, ETH Zurich - Global Health Engineering, Zurich, CH.
- Support faculty, PhD and MSc students in applying reproducible workflows in their research.
- Oversee reproducibility and open science aspects of published journal papers; manage online and social media channels.
- Set up organisational work, task and project management tools.

Since 2018 — **Director & Founder**, Lars Schöbitz GmbH, Zurich, CH.
- Teaching Data Science with R.
- Consultant for city-wide faecal sludge management systems.
- Supporting organisations in establishing strong monitoring and evaluation frameworks.

2021 — **Data Science Instructor**, Statistical Office of the Canton of Zurich, Zurich, CH.
- Teaching a Data Science with R course to civil servants.

2021 — **Data Science Instructor**, University of Colorado Boulder, Boulder, USA.
- Developed and taught an online module to teach R.

2018 to 2021 — **Data and Impact Expert**, Aquaconsult Limited, Essex, UK.
- Venture Advisor for the Transition to Scale Program in Urban Sanitation by Grand Challenges Canada.
- Support to Clean Team (Ghana), Eau&Vie (Bangladesh) and Safi Sana (Ghana).

2020 — **Data Science Intern**, Statistical Office of the Canton of Zurich, Zurich, CH.
- Developed an R Shiny dashboard for analytics of incoming data analysis requests and an R package for web analytics with Matomo.
- Published a fully reproducible news article; supported COVID-19 monitoring and data analysis; organised a session at DINAcon 2020.

2020 — **Consultant**, Eawag / Sandec, Duebendorf, CH.
- Supported the preparation of factsheets for a publication on city-wide inclusive sanitation (CWIS) approaches and tools.

2019 to 2020 — **Lead Consultant**, Inter-American Development Bank, Cap-Haitien, Haiti.
- Led a team of consultants developing city-wide faecal sludge management systems.
- Detailed review of a design proposal for a faecal sludge treatment plant; preparation of a Shit Flow Diagram (SFD) for Cap-Haitien.

2017 to 2020 — **Data Scientist**, Biomass Controls PBC, Putnam, USA.
- Building tools for data visualisation and business intelligence using the R Shiny framework.
- Support of R&D work and preparation of results for publications and conferences.

2018 — **Data Scientist**, Gather, London, UK.
- Supported conceptualization of a data dive for urban sanitation data.
- Cleaned and prepared toilet census data (Kanyama compound, Lusaka, Zambia) for analysis.

2016 to 2018 — **Consultant**, World Health Organization, Geneva, CH.
- Global monitoring of the sanitation chain: proof of concept testing for monitoring indicators on safely managed sanitation (SDG 6.2) and safely treated wastewater (SDG 6.3).
- Production of country files for SDG 6.3.1 (proportion of wastewater safely treated).

2012 to 2016 — **Research Project Manager**, Eawag / Sandec, Duebendorf, CH.
- Project management with global, multi-disciplinary teams (RRR Business Models, SFD Methods and Tools development).
- Research lead on projects developing diagnostic tools for faecal sludge management.
- Mentor, lecturer and networker.

2011 to 2012 **Research Assistant**, PRG, University of KwaZulu-Natal, Durban, ZA.
- Collected water samples from a DEWATS plant; emptied pit latrines; built a rig to separate pit contents into constituents for analysis.

EDUCATION

2019 to 2020 **RStudio Instructor Programme**, RStudio (Posit).
- Certified as an RStudio (Posit) Instructor.

2008 to 2012 **Environmental and Hygiene Engineering (Dipl.-Ing.)**, Mittelhessen University of Applied Sciences (THM), Giessen, DE.
- Thesis: Performance Evaluation of a full-scale DEWATS plant in South Africa.

2010 **ERASMUS Exchange**, Umea University, Umea, SE.
- Main subject: Northern Aquatic Systems, including sampling of surface water and analysis of DOC, COD, main nutrients and heavy metals.

2009 **Change and Project Management**, Mikkeli University of Applied Sciences, Mikkeli, FI.

OPEN DATA AND SOFTWARE

Since 2026 **AGUASAN Workshop 2027 - Publishing WASH Data Together**, AGUASAN Community of Practice / SDC Water Network.
- Author of a topic proposal for a five-day working retreat where WASH-sector participants publish datasets while learning Git/GitHub, Quarto and safe use of AI agents; bridges academia (ETH), the WASH community and Swiss government.

Since 2026 **dataitd24 - data from the FAIRqual workshop at ITD24**, Maintainer.
- R data package with the data of the FAIRqual workshop at the ITD24 conference (Utrecht, November 2024): digitised flipcharts, a qualitative codebook with 50 codes and the workshop documentation; v1.0.0 released in February 2026.
- Includes an article comparing the GitHub to Zenodo and the SWISSUbase publishing workflows.
- DOI: https://doi.org/10.5281/zenodo.18493525

Since 2026 **Qualitative Data Governance for Transdisciplinary and Participatory Research - e-learning module**, Co-author.
- Open e-learning module on the ETH Domain Data Management Campus (ETH Board ORD Program, Measure 3 Open Educational Resources), published in September 2026. Contributed to the unit on levels of access and data repositories (Unit 4.3.2).
- Mohr, F., Marqués, L., Reifenrath, J., Schöbitz, L., Simon, L., & Chapman, M.
- DOI: https://doi.org/10.5281/zenodo.23016356

Since 2025 **fairenough - tidy data into documented R packages**, Author.
- Converts tidy data into R packages with documentation websites and AI-assisted descriptions.

Since 2024 **jmpwashdata - WHO/UNICEF JMP data**, Author.
- R package providing tidy access to WHO/UNICEF Joint Monitoring Programme WASH data.

Since 2023 **Data Science for openwashdata (ds4owd)**, Course Creator & Instructor.
- Designed and taught two free online cohorts on literate programming with Quarto and R; hundreds of participants across many countries.
- Focus on data management, reproducibility and publishing open data.

Since 2023 **washr - R package for publishing openwashdata resources**, Author.
- An R package that makes publishing openwashdata data resources easier.

Since 2022 **openwashdata community**, Founder & Community Lead.
- Built and lead a global open-data community for the WASH sector; 90+ open datasets published as documented R data packages.
- Created the {washr} and {fairenough} R packages that turn tidy data into FAIR, documented data packages.
- See openwashdata.org

Since 2021 **rstatsZH - open course materials & infrastructure**, Creator & Maintainer.
- Built and maintain the open, reusable course materials, website templates and Git-based infrastructure behind the rstatsZH training programme (course delivery is under Teaching).

TEACHING

2024 to 2026 **rstatsZH - Data Science mit R (redeveloped, k009-k012)**, Statistical Office of the Canton of Zurich, Zurich, CH.

2022 to 2026 **CVEN 5837/5873/5999 - Data Analytics for Development (4 iterations)**, University of Colorado Boulder, Boulder, USA (online).

September to December 2025 **Data Science for openwashdata (ds4owd), Cohort 2**, openwashdata / ETH Zurich.

2022 to 2025 **Research Beyond the Lab - Open Science and Research Methods (3 iterations)**, ETH Zurich, Zurich, CH.

April 2024 **Quarto - Authoring and Publishing for Collaborative Scientific Writing**, ETH Zurich - Global Health Engineering.

October 2023 to February 2024	**Data Science for openwashdata (ds4owd), Cohort 1**, openwashdata / ETH Zurich.
October 2023	**Open Science Tools for Collaborative Scientific Writing**, ETH Zurich - Global Health Engineering, Zurich, CH.
2021 to 2023	**rstatsZH - Data Science mit R (k001-k008)**, Statistical Office of the Canton of Zurich, Zurich, CH.
2017	**Lecturer - Faecal Sludge Management**, Mittelhessen University of Applied Sciences (THM), Giessen, DE.
2016	**Lecturer - Sanitation MOOC**, Massive Open Online Course: Planning & Design of Sanitation Systems and Technologies.
2013 to 2016	**Lecturer - Faecal Sludge Management**, NADEL (ETH Zurich) & EPF Lausanne, Switzerland.

WORKSHOPS

20 November 2026	**Git for Scientists: hands-on workshop**, Swiss Research Data Conference (RDC 2026) - SRDSN, Bern, CH. repository
September 2026	**Agents for Scientists**, ETH Zurich - Global Health Engineering, Zurich, CH.
July 2026	**Git for Scientists**, ETH Zurich - Global Health Engineering, Zurich, CH.
June 2026	**Quarto for Collaborative Scientific Writing - RDM Summer School**, ETH Zurich, Zurich, CH.
May 2025	**Git for Scientists**, CIS, ETH Zurich, Zurich, CH.
November 2024	**FAIR Data Publishing**, UKZN WASH Centre, UKZN, South Africa.
June 2024	**Quarto for Collaborative Scientific Writing - RDM Summer School**, ETH Zurich, Zurich, CH.
November 2023	**Data Stewardship - SST Educational Workshop**, Swiss Toxicology Society (SST), Basel.
October 2023	**Reference Management with Zotero**, Hubris to Hope (HS23).
June 2023	**open by default - a data management strategy**, ETH Research Data Management Summer School, Zurich, CH.
February 2023	**Data Stewardship Network Kick-off**, ETH Zurich, Zurich, CH.
2021	**Data Science for WASH - An opportunity for the sector**, Colorado WASH Symposium, online.

TALKS

30 November 2026	**Teaching Git, GitHub, and coding agents to researchers who do not come from IT**, EnhanceR Symposium 2026 - EnhanceR association, Villigen, CH. repository
19 November 2026	**Embedded data stewardship and research intelligence: five years of practice at a research group at ETH Zurich**, Swiss Research Data Conference (RDC 2026) - SRDSN, Bern, CH. repository
September 2026	**Git for Scientists - so that we can still prove what is human**, Data Stewardship Network (DSN) Autumn Community Meeting, ETH Zurich, Zurich, CH.
August 2026	**Data repositories and access restrictions - FAIRqual stakeholder workshop**, FAIRqual stakeholder workshop with the Reallabor Jurapark, ETH Zurich, Zurich, CH.
May 2026	**Hello, I am a Data Steward - ReproducibiliTea Journal Club**, University of Zurich.
September 2025	**Publishing Open Metadata for ORD Projects of the ETH Domain**, Swiss Open Academic Data (SOAD) Day.
June 2025	**Hello, I am a Data Steward - Career Talk**, Research Software Engineering and Data Stewardship Career Panel.
June 2025	**FAIRqual - FAIR Data Practices for Qualitative Research in Transdisciplinarity**, Advancing Open Research Data (ORD) Practices: collaboration and innovation in the ETH Domain (ETH Domain ORD Day), Bern, CH.
May 2025	**Computational Reproducibility Seminar**, Swiss Reproducibility Network (SwissRN).
May 2025	**Plan for tomorrow today - D-USYS Transdisciplinary Lab Brown Bag**, ETH Zurich D-USYS TdLab.
May 2025	**Open Research Data Program of the ETH Board - open metadata dataset**, ETH ORD Programme.
November 2024	**Plan for tomorrow today - a model for data stewardship**, ETH Library 17:15 Kolloquium.
October 2024	**Swiss Research Data Day (SRDD)**, SRDD.
September 2024	**Data Stewardship at Global Health Engineering**, ETH Zurich.
June 2024	**Plan for tomorrow today - why you need a data steward**, Swiss Reproducibility Conference (SwissRN).
March 2021	**Berufsbild EZ Seminar**, Albert-Ludwigs-University of Freiburg, Freiburg, DE.
2021	**Die datengetriebene Verwaltung: Daten- und Statistikkompetenz für den Kanton Zürich**, Conference on Digital Sustainability (DINAcon).
2020	**DINAcon 2020**, Conference on Digital Sustainability (DINAcon), online.
2020	**UN World Data Forum 2020**, United Nations World Data Forum, online.
2019	**5th Faecal Sludge Management Conference**, FSM5, Cape Town, ZA.
2016	**UNC Water and Health Conference**, UNC, Chapel Hill, USA.
2016	**Sandec Research Seminar**, Eawag / Sandec, Duebendorf, CH.
2015	**UNC Water and Health Conference**, UNC, Chapel Hill, USA.
2015	**Stockholm World Water Week**, SIWI, Stockholm, SE.
2015	**4th AfricaSan Conference**, AfricaSan, Dakar, SN.
2015	**3rd Faecal Sludge Management Conference**, FSM3, Hanoi, VN.
2014	**37th WEDC International Conference**, WEDC, Hanoi, VN.
2014	**Stockholm World Water Week**, SIWI, Stockholm, SE.

2013 **36th WEDC International Conference**, WEDC, Nakuru, KE.
2012 **2nd Faecal Sludge Management Conference**, FSM2, Durban, ZA.
2012 **Water Institute of Southern Africa (WISA) Conference**, WISA, Cape Town, ZA.

BLOG AND WRITING

20 October 2026 **From a raw spreadsheet to a published data package in a day**, openwashdata Blog.
July 2026 **Who wrote this? Scientific integrity at ETH Zurich in the age of AI**, GHE Blog.
22 June 2026 **From course project to published dataset and paper: a collaboration with Kelsey Shaw**, openwashdata Blog.
10 December 2024 **openwashdata hackathon**, openwashdata Blog.
6 August 2024 **openwashdata phase 2 proposal funded**, openwashdata Blog.
2 April 2024 **Strategic partnership with SOIL**, openwashdata Blog.
March 2024 **Plan for tomorrow today: why you need a data steward**, GHE Blog (with E. Tilley).
February 2024 **Motivation for Team Award**, GHE Blog.
February 2024 **Attention, Prof: You need a data steward for your team**, GHE Blog.

PUBLICATIONS

2025 **Light absorbing carbonaceous aerosols: Concentrations and sources in an under-monitored region**, Ecotoxicology and Environmental Safety.
- Co-author. Peer-reviewed journal article.

2020 **Estimating Safely Managed Sanitation in Urban Areas; Lessons Learned From a Global Implementation of Excreta-Flow Diagrams**, Frontiers in Environmental Science.
- Peer-reviewed journal article on SDG 6.2 monitoring.

2019 **Towards an off-grid fecal sludge treatment unit: demonstrating energy positive thermal treatment**, Gates Open Research.
- Peer-reviewed journal article.

2018 **Methods to reliably estimate faecal sludge quantities and qualities for the design of treatment technologies and management solutions**, Journal of Environmental Management.
- Peer-reviewed journal article.

2017 **GIS Analysis and Optimisation of Faecal Sludge Logistics at City-Wide Scale in Kampala, Uganda**, Sustainability.
- Peer-reviewed journal article.

2014 **FAQ: Faecal sludge quantification and characterization - field trial of methodology in Kampala, Uganda**, Sandec News.
- Journal article.

CONFERENCE PAPERS

2018 **Remote Sensing, Mobile Applications and Open Data Science Tools for Better Monitoring of Sanitation Systems**, HIC 2018, 13th International Conference on Hydroinformatics.
- Conference paper.

2017 **Market driven approach for faecal sludge treatment products**, 40th WEDC International Conference.
- Co-author. Conference paper.

2014 **FAQ: Faecal Sludge Quantification and Characterization - field trial of methodology in Hanoi, Vietnam**, 37th WEDC International Conference.
- Conference paper.

TECHNICAL REPORTS

2015 to 2016 **SFD Promotion Initiative - Final Reports (7 cities)**, Sustainable Sanitation Alliance / Eawag-Sandec.
- Lead / co-author of Shit Flow Diagram (SFD) reports for Bignona (SN), Hanoi & Danang (VN), Kampala (UG), Dar es Salaam (TZ), Khulna (BD), Nonthaburi (TH).

2014 **RRR-Project: From Research to Implementation, Component 1 - Waste Supply and Availability**, Eawag-Sandec.
- Technical report on resource recovery and reuse.

GRANTS AND FUNDING

2024 to 2026 **openwashdata - establishing Data Stewards and increasing FAIRness**, ETH Board ORD Program (Explore, 2nd call).
- Wrote the proposal and managed the project (Prof. Elizabeth Tilley, PI). Placed funded data stewards in partner organisations (UKZN, South Africa; BASEflow, Malawi) and delivered a 12-module data stewardship training programme.

2024 to 2026 **FAIRqual - FAIR Data Practices for Qualitative Research in Transdisciplinarity**, ETH Board ORD Program (Explore, Measure 1).

- Co-wrote the proposal and contributed to the project (team across TdLab and Global Health Engineering, ETH Zurich). Developed tools and standards for sharing hard-to-anonymise qualitative research data following FAIR principles.

2024 to 2025 **washcollab - Enhancing Global WASH Data Accessibility through Collaborative Initiatives**, ETH Board ORD Program (Contribute).

- Wrote the proposal and managed the project (Prof. Elizabeth Tilley, PI). A partnership between WASHWeb and openwashdata that maintained open-source tools for authoritative WASH data (WHO/UNICEF JMP, World Bank, UN-Water GLAAS, OECD).

2023 to 2024 **openwashdata - Open WASH data by building Open Science Competencies and Community**, ETH Board ORD Program (Explore, 1st call).

- Wrote the proposal and managed the project (Prof. Elizabeth Tilley, PI). Built the infrastructure, community and FAIR data publishing workflow for open WASH data; 18 months.

2023 to 2024 **openjmp - Open JMP: unlocking the potential of global indicator data**, ETH Board ORD Program (Contribute).

- Wrote the proposal and managed the project (Prof. Elizabeth Tilley, PI). In partnership with the WHO/UNICEF Joint Monitoring Programme, turns decades of WASH coverage data into documented open R packages; 15 months.

PROPOSALS

Submitted 2026 **SDSC openwashdata - Reviewed, open data publishing for the WASH sector, end to end at pilot scale**, Swiss Data Science Center (SDSC), National Call 2026 (Collaboration and Expertise Request).

- Wrote the proposal as PI (Prof. Elizabeth Tilley, co-PI). Requests six months of SDSC engineering time, as a Proof of Concept, to turn the openwashdata publishing pipeline into a service on secure ETH infrastructure that takes submitted files to a reviewed, citable data package.

Submitted 2026 **FAIR by doing - A hands-on programme that trains ETH researchers to make their own research data FAIR and to declare their use of AI tools**, ETH FAIR Competence Funding (2nd call).

- Wrote the proposal as corresponding applicant, with Adriana Clavijo Daza. A twelve-month programme from 1 February 2027 in which fifteen ETH researchers, mostly postdoctoral researchers and doctoral students, take a dataset of their own to a published, citable data package and declare their use of AI tools.

Not funded 2025 **Swiss-Ghana partnership - Building Computational Research Competencies: A Swiss-Ghana Partnership for Open Science Capacity Development**, Swiss Leading House Africa, Research Partnership Grant 2025-2028 (SERI, Swiss TPH with the University of Basel).

- Wrote the proposal (Prof. Elizabeth Tilley, Swiss PI; Dr. Eugene Appiah-Effah, KNUST, Ghana). A ten-month programme with a one-week bootcamp at ETH Zurich for eight KNUST researchers from the ds4owd course, followed by remote mentorship in Open Science workflows.

AWARDS

2024 **National Prize for Open Research Data 2024**, Swiss Academies of Arts and Sciences.

- Awarded to openwashdata, the global open-data community for the WASH sector (project led by Prof. Elizabeth Tilley, Global Health Engineering, ETH Zurich).
- Recognition for advancing FAIR and Open Research Data practices in the water, sanitation and hygiene sector.
- Watch the prize video: youtu.be/590qPtQNCD0

PROFESSIONAL ROLES

Since 2026 **Co-Lead, AI Disclosure Working Group, Global Federation of Reproducibility Networks (GFRN)**, Global Federation of Reproducibility Networks (GFRN).

- Co-lead in the GFRN AI Disclosure Working Group, contributing the reproducibility community's perspective to the Global Reporting Standard for AI Disclosure in Research (a joint ISC, COPE, STM, and GYA initiative). The GFRN connection runs through the ETH Zurich SwissRN local node.
- Co-lead of three of the six topics since September 2026: taxonomy of AI use, responsibility and accountability, and the cross-cutting review of existing reporting standards and literature.

Since 2025 **Local Node Leader, Swiss Reproducibility Network (SwissRN)**, ETH Zurich - Global Health Engineering.

- Local Node Leader for ETH Zurich in SwissRN, promoting reproducible and open research practices across the institution.

Since 2023 **Member, Data Stewardship Network (DSN)**, ETH Zurich.

- Data steward for the Global Health Engineering group in ETH Zurich's Data Stewardship Network, an ETH Library / swissuniversities ORD initiative embedding Open Research Data practices across research groups.

2025 to 2026 **Member, Expert Sounding Group - ORD Career Paths (Measure 5)**, ETH Board ORD Program.

- Expert sounding group member for the Measure 5 project Career Paths for Open Research Data Professionals, advising the project through planning, progress monitoring and validation of the final report.